

SAT-Style Practice Test - December 2024 Original Practice Form

All passages, questions, names, numbers, data, and answer choices in this file are newly written for SAT-style practice. Answers and math solutions are included for checking.

SECTION 1, MODULE 1: Reading and Writing

Question 1

Skill: Words in Context

The Harbor Craft Fellowship was established to recognize makers whose work preserves regional traditions while expanding them in original ways. One of its first recipients was ceramicist Nura Selim, who was honored for her lifelong _____ the art of hand-thrown tile making.

Which choice completes the text with the most logical and precise word or phrase?

- A) dedication to
- B) uncertainty about
- C) withdrawal from
- D) misinterpretation of

Answer: A

Question 2

Skill: Words in Context

The following text is adapted from a nineteenth-century novel. As Mira crossed the courtyard, she kept glancing toward the gate, eagerly anticipating the arrival of the traveling bookseller whose wagon appeared in town only twice a year.

As used in the text, what does the word "anticipating" most nearly mean?

- A) Predicting
- B) Avoiding
- C) Looking forward to
- D) Explaining

Answer: C

Question 3

Skill: Words in Context

The city archive recently began a crowdsourced project in which volunteers transcribe handwritten shipping logs from the 1880s. The volunteers then add brief notes that help researchers identify _____ between merchants, ship captains, and ports.

Which choice completes the text with the most logical and precise word or phrase?

- A) connections
- B) apologies
- C) restrictions
- D) contradictions

Answer: A

Question 4

Skill: Words in Context

The Sundar River delta is a remarkably _____ landscape: its channels widen, split, and disappear as monsoon floods deposit fresh layers of silt.

Which choice completes the text with the most logical and precise word or phrase?

- A) changeable
- B) permanent
- C) isolated
- D) uniform

Answer: A

Question 5

Skill: Central Ideas and Details

Tired of throwing away coffee grounds? One practical response is to use them in the garden. When mixed carefully into compost, spent grounds can add nitrogen to soil and may help some plants grow. Gardeners should avoid piling grounds directly around stems, however, because dense clumps can block water and air.

Which choice best states the main purpose of the text?

- A) To present a useful but limited way to reuse a common waste product
- B) To compare several commercial fertilizers used by gardeners
- C) To argue that coffee grounds should replace all other compost materials
- D) To describe the flavor differences among types of coffee

Answer: A

Question 6

Skill: Text Structure and Purpose

Early detective stories often relied on puzzles that were solved by a brilliant outsider. Recent crime fiction, however, frequently shifts attention from a single investigator to the social conditions surrounding a crime. Novelist Jae-min Park, for example, uses the mystery form to examine housing policy and neighborhood change.

Which choice best describes the overall structure of the text?

- A) It describes a feature of earlier fiction, then explains how some recent fiction differs from it.
- B) It criticizes one novelist, then praises a different novelist for the same quality.
- C) It lists several causes of a historical event, then ranks those causes by importance.
- D) It introduces a scientific hypothesis, then describes an experiment testing it.

Answer: A

Question 7

Skill: Function of a Sentence

In a 2018 survey, Lina Ortiz and her team reported that rooftop gardens in one neighborhood reduced indoor temperatures by an average of 1.8 degrees Celsius. A later review by architect Musa Rahman cautioned that the survey included only ten buildings, so its results may not apply to all urban settings. Such limited samples can make the effects of design interventions seem more general than they are.

Which choice best describes the function of the underlined sentence in the text?

- A) It states a finding that the text later qualifies.
- B) It gives a definition that is used throughout the text.
- C) It presents an opposing view that the author rejects as false.
- D) It describes the method used in a follow-up experiment.

Answer: A

Question 8

Skill: Inferences

Testing equipment in polar regions is difficult because batteries, seals, and moving parts can fail in extreme cold. To prepare instruments for Antarctic fieldwork, engineers use environmental chambers that imitate low temperatures and dry winds. By observing devices in these chambers, engineers can identify weaknesses before the equipment is shipped south.

Based on the text, what is one reason environmental chambers are valuable to engineers?

- A) They allow engineers to test whether equipment will function under harsh conditions.
- B) They show engineers how Antarctic winds have changed over many decades.
- C) They replace the need to transport any equipment to the field.
- D) They compare the chemical composition of Antarctic ice with that of other ice.

Answer: A

Question 9

Skill: Command of Evidence: Quantitative

[Table: Millions of metric tons of rice produced in 2005 and 2020. Country | 2005 | 2020. Vietnam | 35.8 | 42.7. Thailand | 28.0 | 30.2. Bangladesh | 39.8 | 54.9. Cambodia | 4.2 | 10.9.] While researching rice production, a student notes that Cambodia produced 4.2 million metric tons of rice in 2005 and ____

Which choice most effectively uses data from the table to complete the statement?

- A) 10.9 million metric tons in 2020.
- B) 54.9 million metric tons in 2020.
- C) 30.2 million metric tons in 2020.
- D) 35.8 million metric tons in 2020.

Answer: A

Question 10

Skill: Command of Evidence

Historian Elena Bae argues that a notation system used by a fifteenth-century mapmaker was an early method for marking fractional distances. She claims that the small diagonal marks in the mapmaker's tables were not decorative symbols but represented portions of a unit.

Which finding, if true, would most directly support Bae's claim?

- A) A navigation manual copied by the same mapmaker explains that diagonal marks in distance tables indicate parts of a unit.
- B) Several manuscripts from the same period contain decorative borders with diagonal marks.
- C) The mapmaker sometimes used red ink to highlight city names on coastal maps.
- D) Some of the mapmaker's maps are inaccurate because coastlines were difficult to survey.

Answer: A

Question 11

Skill: Command of Evidence: Quantitative

[Table: Average number of minutes two groups spent completing a training task after programs began. Program | 4 weeks | 6 months. Video tutorial | 46 | 29. Peer coaching | 45 | 41.] Researchers concluded that video tutorials were more effective than peer coaching at reducing the time needed to complete the task, though the effect took months to appear.

Which choice best uses data from the table to strengthen the researchers' conclusion?

- A) The groups had nearly identical averages at 4 weeks, but by 6 months the video-tutorial group averaged far less time than the peer-coaching group.
- B) Both groups needed more time at 6 months than at 4 weeks.
- C) The peer-coaching group was faster than the video-tutorial group at both measured points.
- D) The video-tutorial group and peer-coaching group showed no change over time.

Answer: A

Question 12

Skill: Command of Evidence: Graphical

[Graph: Four poems were rated on a 1-to-8 enjoyment scale by readers who had and had not been told the poem's final image in advance. For "Harbor Bell," ratings were 5.0 unspoiled and 5.1 spoiled. For "Winter Orchard," 6.0 unspoiled and 4.8 spoiled. For "Glass River," 4.2 unspoiled and 7.0 spoiled. For "North Window," 5.5 unspoiled and 5.4 spoiled.] Researchers found that advance knowledge affected enjoyment differently depending on the poem. The largest positive difference for spoiled readers occurred for ____

Which choice most effectively uses data from the graph to complete the statement?

- A) "Glass River."
- B) "Winter Orchard."
- C) "Harbor Bell."
- D) "North Window."

Answer: A

Question 13

Skill: Inferences

Most people are right-handed, but studies of tool use among wild capuchin monkeys show no strong preference for one hand. A laboratory study, however, found that captive capuchins appeared mostly right-handed when solving a puzzle box. The researchers noted that captive monkeys had been trained on a box whose latch was easiest to pull with the right hand.

Which choice most logically completes the text?

- A) the captive monkeys' apparent hand preference may have resulted partly from the design of the task.
- B) the wild monkeys probably would have preferred their right hands if they had been observed longer.
- C) capuchin monkeys are more strongly right-handed than humans are.
- D) the puzzle box could not have influenced the monkeys' behavior in any way.

Answer: A

Question 14

Skill: Inferences

Two species of marsh frogs choose breeding pools based on water depth and the number of reeds around the pool. Researchers found that one species prefers pools with dense reeds, while both species avoid pools that dry up in less than thirty days. The researchers therefore concluded that neither species is likely to choose a pool where ____

Which choice most logically completes the text?

- A) standing water lasts for fewer than thirty days.
- B) dense reeds are present throughout the spring.
- C) another species of frog has never been observed.
- D) the water is deeper than usual.

Answer: A

Question 15

Skill: Quantitative Reasoning

It is difficult for publishers to balance readability, durability, and cost in very long books. A 960-page edition of Novel A contains about 480,000 words, while a 1,200-page edition of Novel B contains about 600,000 words.

Which choice most logically completes the text?

- A) the two editions have the same average number of words per page.
- B) Novel A has more total words than Novel B.
- C) the edition of Novel B has fewer pages than the edition of Novel A.
- D) page count alone proves that Novel B takes twice as long to read as Novel A.

Answer: A

Question 16

Skill: Standard English Conventions: Verb Form

Portrait of Admiral Leona Voss is one of the few remaining paintings by nineteenth-century artist Clara Wen. ____ in 1842, the portrait is now held by the Eastport Museum.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) Completed
- B) Completes
- C) Will complete
- D) Was completing

Answer: A

Question 17

Skill: Standard English Conventions: Form, Structure, and Sense

Artificial coral blocks can help rebuild damaged reefs, a function well suited for coastal restoration projects, ____ from the porous structure of the blocks.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) stemming
- B) stems
- C) to stem
- D) having stemmed

Answer: A

Question 18

Skill: Standard English Conventions: Boundaries

The Tanoan language family includes several languages spoken in the American Southwest, including Tiwa and Tewa, ____ and it remains an important focus for linguists such as Maria Yazzie.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) languages,
- B) languages;
- C) languages
- D) languages:

Answer: B

Question 19

Skill: Standard English Conventions: Boundaries

For millions of years, Earth's climate has shifted between colder and warmer intervals. This has not always been the ____, scientists can reconstruct many of those shifts from ice cores.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) case, though
- B) case though
- C) case. Though
- D) case, though,

Answer: A

Question 20

Skill: Standard English Conventions: Punctuation

Artist Mauro Kenji turns ordinary tools into playful sculptures. In one installation, a row of spoons becomes a flock of birds hanging from a ____ their bowls become wings, and their handles become tails.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) wire:
- B) wire,
- C) wire;
- D) wire

Answer: A

Question 21

Skill: Transitions

Twentieth-century architects often celebrated glass towers as symbols of technological progress. Architect Amina Kroll, however, preferred thick clay walls and shaded courtyards. ____ her buildings show that modern design can draw from older regional forms.

Which choice completes the text with the most logical transition?

- A) Thus,
- B) Nevertheless,
- C) For instance,
- D) Instead,

Answer: A

Question 22

Skill: Transitions

In Country X, citizens must be 18 years old to vote in national elections. ____ citizens in Country Y may vote at 16.

Which choice completes the text with the most logical transition?

- A) By contrast,
- B) Therefore,
- C) For example,
- D) In addition,

Answer: A

Question 23

Skill: Transitions

Sleek trains are usually more aerodynamic than boxy freight cars. ____ a train with a rounded front experiences less air resistance than a train with a flat front.

Which choice completes the text with the most logical transition?

- A) Specifically,
- B) On the other hand,
- C) Nevertheless,
- D) In conclusion,

Answer: A

Question 24

Skill: Transitions

As a supporter of a strong central government, writer Nathaniel Pike praised the new constitution. ____ in a series of essays, he answered critics who feared that the document gave too much power to the national legislature.

Which choice completes the text with the most logical transition?

- A) Furthermore,
- B) By comparison,
- C) Nevertheless,
- D) In other words,

Answer: A

Question 25

Skill: Expression of Ideas: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- Glenport is a town in western Oregon.
- Town governments provide services to residents.
- One service Glenport provides is public library funding.
- Glenport covers about 18 square miles.
- Oregon has 241 incorporated towns and cities.

The student wants to emphasize the size of Glenport. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Glenport, a town in western Oregon, covers about 18 square miles.
- B) Town governments such as Glenport's provide services to residents.
- C) Public library funding is one of the services provided by Glenport.
- D) Oregon has 241 incorporated towns and cities, including Glenport.

Answer: A

Question 26

Skill: Expression of Ideas: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- The country has more than 120 protected coastal reserves.
- Blue Bay Reserve is a 3,400-acre reserve in Maine.
- Blue Bay Reserve was created to protect nesting terns.
- Sand Point Reserve is a 210-acre reserve in California.
- Sand Point Reserve was created to protect nesting plovers.

The student wants to emphasize a similarity between Blue Bay Reserve and Sand Point Reserve. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Both Blue Bay Reserve and Sand Point Reserve were created to protect nesting birds.
- B) Blue Bay Reserve is larger than Sand Point Reserve.
- C) The country has more than 120 protected coastal reserves.
- D) Blue Bay Reserve is in Maine, whereas Sand Point Reserve is in California.

Answer: A

Question 27

Skill: Expression of Ideas: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- Painter Sofia Lenz creates surrealist collages.
- Surrealist collage often uses irrational juxtaposition.
- Lenz's work *Doorway at Noon* includes a fish, a staircase, and a thundercloud in the same room.
- Lenz says her art explores the logic of dreams.

The student wants to provide a specific example of irrational juxtaposition in Lenz's art. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) *Doorway at Noon* places a fish, a staircase, and a thundercloud in the same room.
- B) Sofia Lenz is a painter who creates surrealist collages.
- C) Surrealist collage often uses irrational juxtaposition.
- D) Lenz says her art explores the logic of dreams.

Answer: A

SECTION 1, MODULE 2: Reading and Writing

Question 1

Skill: Words in Context

Like many coastal deltas, the Alder Delta is ____; storms and tides continually reshape its narrow islands and channels.

Which choice completes the text with the most logical and precise word or phrase?

- A) dynamic
- B) unrivaled
- C) sustainable
- D) immutable

Answer: A

Question 2

Skill: Words in Context

Art historian Priya Menon argued that two portraits once thought unrelated were actually connected. By comparing pigment layers and facial proportions, Menon ____ her claim that both works came from the same studio.

Which choice completes the text with the most logical and precise word or phrase?

- A) corroborated
- B) diminished
- C) obscured
- D) postponed

Answer: A

Question 3

Skill: Words in Context

Although composers often draw on earlier traditions, their influences can be broad. The music of Rafiq Dar, for example, can be said to ____ elements of classical strings, village percussion, and electronic sampling.

Which choice completes the text with the most logical and precise word or phrase?

- A) incorporate
- B) renounce
- C) foretell
- D) conceal

Answer: A

Question 4

Skill: Words in Context

During a period of market instability, analyst Nola Chen found that emergency lending decisions were not ____ political alliances; instead, they were more strongly associated with the condition of each country's banking system.

Which choice completes the text with the most logical and precise word or phrase?

- A) predicated on
- B) detached from
- C) complicated by
- D) irrelevant to

Answer: A

Question 5

Skill: Central Ideas and Details

Though poet Marian Bell is less famous than several of her contemporaries, critics have increasingly praised her work. One reviewer notes that Bell's poems combine plainspoken scenes with philosophical reflection. Rather than choosing between realism and abstraction, Bell shows that a poem can do both.

Which choice best states the main purpose of the text?

- A) To present a reason why critics value a writer's work
- B) To explain why a writer stopped publishing poetry
- C) To compare a poet with a novelist who admired her
- D) To argue that all poems must fit into one of two categories

Answer: A

Question 6

Skill: Text Structure and Purpose

In the village, Dr. Halden was widely considered generous and wise. He gave calm advice, visited sick neighbors, and spoke modestly about his accomplishments. Yet he also made sure that every public notice mentioned his name, and he privately complained whenever praise went to someone else.

Which choice best describes the overall structure of the text?

- A) It presents the favorable opinion people have of Dr. Halden and then reveals behavior that complicates that opinion.
- B) It describes a conflict between two villages and then explains how Dr. Halden resolved it.
- C) It explains why Dr. Halden left his village and then gives evidence that he returned.
- D) It lists Dr. Halden's failures and then argues that his neighbors never respected him.

Answer: A

Question 7

Skill: Cross-Text Connections

Text 1: In the novel *The Hill Station*, the city train terminal represents an entrance to modern life. Its steel roof and crowded platforms suggest movement, ambition, and the possibility of leaving old expectations behind.

Text 2: Critics of *The Hill Station* often note that the novel contrasts the modern city with the quiet mountain village. The terminal, where the protagonist begins her journey, connects these two worlds but also highlights the tension between them.

Based on the texts, the author of Text 2 would most likely agree with which statement about the terminal as it is depicted in Text 1?

- A) It links the protagonist's city life to the mountain village while also emphasizing the contrast between the two settings.
- B) It is included only to show that the protagonist dislikes train travel.
- C) It represents the novel's rejection of modern ambition in favor of complete isolation.
- D) It has no symbolic importance beyond identifying where the journey begins.

Answer: A

Question 8

Skill: Cross-Text Connections

Text 1: A recent model predicts that warming lakes will reduce habitat for silver minnows but may not greatly affect stone perch, which tolerate warmer water.

Text 2: Conservation agencies cannot address every environmental threat at once. They should use the best available data to decide which species are most at risk and should receive the most immediate attention.

Based on the texts, both authors would most likely agree with which statement?

- A) Agencies should prioritize conservation efforts for silver minnows over stone perch if the model is accurate.
- B) Stone perch will require more urgent protection than silver minnows in all lakes.
- C) No species in warming lakes can benefit from conservation planning.
- D) Conservation decisions should ignore projected habitat changes.

Answer: A

Question 9

Skill: Inferences

The following text is adapted from a novel. Whenever guests lingered in the front room after supper, Lena found reasons to open windows, gather empty cups, and mention the lateness of the hour. The front room, she knew, would become her classroom at dawn, and she needed the tables clear before the children arrived.

Based on the text, what most likely motivates Lena's behavior?

- A) She wants the guests to leave so she can prepare the room for teaching.
- B) She is trying to persuade the guests to enroll their children in school.
- C) She dislikes all visitors and refuses to speak with them.
- D) She needs the guests to help move the tables into another building.

Answer: A

Question 10

Skill: Command of Evidence

Edinburgh has installed hardened shoreline structures, such as seawalls and stone barriers, along much of its coast. Ecologist Dana Pritchard surveyed bird communities at several coastal sites and found that seawalls were more strongly associated with lower bird-community integrity scores than natural rock barriers were.

Which finding, if true, would most directly illustrate Pritchard's finding?

- A) A site with a high percentage of seawall had lower bird-community integrity scores than a site with a high percentage of natural rock barrier.
- B) A site with no shoreline structures had the same score as a site with no birds.
- C) Two sites with equal percentages of natural rock barrier had different numbers of visitors.
- D) The difference between two high-seawall sites was not statistically significant.

Answer: A

Question 11

Skill: Command of Evidence: Quantitative

[Table: Effect of neighboring plants on visits to target flowers. Neighboring species | Target species | Effect value. Silver thistle | meadow lupine | 0.318. Red clover | dune aster | -0.527. Saltbush | blue flax | -0.211. White sage | desert mallow | 0.446. Negative values mean the neighboring species reduced visits to the target species.] Based on the table, the two neighboring species with harmful effects are ____

Which choice most effectively uses data from the table to complete the statement?

- A) red clover and saltbush.
- B) silver thistle and white sage.
- C) red clover and white sage.
- D) saltbush and silver thistle.

Answer: A

Question 12

Skill: Command of Evidence

A student claims that in the short story "The Evening Ticket," a theater performance briefly allows the main character to escape the pressures of daily life.

Which quotation from "The Evening Ticket" would most effectively illustrate the claim?

- A) "For two hours, the lamps and music seemed to lift the city's noise from her shoulders."
- B) "There were vacant chairs near the aisle, though she chose a seat in the back row."
- C) "The play had begun before she entered, and the usher pointed silently toward the door."
- D) "She wore the same gray coat she had worn to work that morning."

Answer: A

Question 13

Skill: Command of Evidence: Weaken

A researcher played a slow lullaby for infants and found that their heart rates decreased and their pupils narrowed during the song. The researcher concluded that the lullaby relaxed the infants. A critic argues instead that the song merely attracted the infants' attention.

Which finding, if true, would most directly weaken the critic's claim?

- A) Independent studies show that pupil narrowing in infants is more closely associated with relaxation than with attention.
- B) The infants had never heard the specific lullaby before the study.
- C) Parents in the study preferred the lullaby to a faster song.
- D) The lullaby was performed on an unfamiliar instrument.

Answer: A

Question 14

Skill: Command of Evidence: Quantitative

[Table: Composition and fracture toughness of four alloys. Alloy 12: cobalt 33%, nickel 33%, chromium 34%, toughness 260. Alloy 8: cobalt 20%, iron 20%, nickel 20%, chromium 20%, manganese 20%, toughness 216. Alloy 31: aluminum 18%, carbon 16%, cobalt 0%, tungsten 66%, toughness 9. Alloy 47: cobalt 25%, nickel 25%, tantalum 25%, tungsten 25%, toughness 5.] A materials scientist predicted that greater cobalt percentage always corresponds to greater fracture toughness. The table does not support this prediction because ____

Which choice most effectively uses data from the table to complete the text?

- A) Alloy 47 has more cobalt than Alloy 8 but much lower fracture toughness.
- B) Alloy 12 and Alloy 8 both contain cobalt.
- C) Alloy 31 contains no cobalt and has low fracture toughness.
- D) Alloy 12 has the highest fracture toughness in the table.

Answer: A

Question 15

Skill: Inferences

The kumu tree is a rare forest plant whose seeds are usually spread by fruit-eating birds. Although two native bird species that once ate kumu fruit are now extinct, several introduced birds have begun eating the fruit and dropping the seeds across the island. Studies confirm that the introduced birds are dispersing viable seeds.

Which choice most logically completes the text?

- A) the introduced birds may help the kumu tree continue spreading to new areas.
- B) the introduced birds must be preventing all native plants from reproducing.
- C) the kumu tree can no longer produce viable seeds.
- D) the extinct birds are still the only species capable of dispersing kumu seeds.

Answer: A

Question 16

Skill: Standard English Conventions: Boundaries

The human body has three main types of muscle: cardiac, smooth, and skeletal. The orbicularis oris is a skeletal muscle - which controls movements of the lips - and it helps with speaking and eating ____

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) tissue.
- B) tissue,
- C) tissue;
- D) tissue:

Answer: A

Question 17

Skill: Standard English Conventions: Possessives

During the Reconstruction era, state legislator Elias Boone was one of the first African ____ to serve in his state's senate.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) Americans
- B) American's
- C) Americans'
- D) American,

Answer: A

Question 18

Skill: Standard English Conventions: Boundaries

Poet A. R. Valdez, born in 1948, wrote in both Spanish and English, ____ her most influential collection, *Desert Letters*, was published in 1986.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) and
- B) and,
- C) but,
- D) but

Answer: A

Question 19

Skill: Standard English Conventions: Punctuation

The programming language FORTRAN, developed in 1957; Lisp, developed in 1958; and BASIC, ____ are all important in the history of computing.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) developed in 1964,
- B) developed, in 1964
- C) developed in 1964;
- D) developed in 1964

Answer: A

Question 20

Skill: Standard English Conventions: Pronouns

Proto-Len is a hypothesized ancestral language of all Len languages, but because ____ descendants are spoken today, scholars can compare them to infer aspects of Proto-Len grammar.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) its
- B) it's
- C) their
- D) they're

Answer: A

Question 21

Skill: Transitions

Middle English was widely spoken in fourteenth-century England. The English language, ____ was rarely used in official legal documents until later centuries.

Which choice completes the text with the most logical transition?

- A) however,
- B) besides,
- C) in fact,
- D) for example,

Answer: A

Question 22

Skill: Transitions

The skateboard trick called a 900 was once considered nearly impossible. ____ several athletes have performed the trick in major competitions since the early 2000s.

Which choice completes the text with the most logical transition?

- A) Nevertheless,
- B) For this reason,
- C) In conclusion,
- D) By comparison,

Answer: A

Question 23

Skill: Transitions

The common raven and the desert raven look very similar from a distance. Size, ____ remains a helpful identifying feature: the common raven is usually larger.

Which choice completes the text with the most logical transition?

- A) however,
- B) instead,
- C) therefore,
- D) indeed,

Answer: A

Question 24

Skill: Expression of Ideas: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- Sculptor Alma Ruiz was born in 1920.
- Ruiz is best known for metal sculptures about migration.
- Harbor Gate is a 1987 sculpture by Ruiz.
- Night Map is a 1993 print by Ruiz.

The student wants to provide an example of one of Ruiz's sculptures. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Harbor Gate is a 1987 sculpture by Alma Ruiz.
- B) Alma Ruiz was born in 1920.
- C) Night Map is a 1993 print by Alma Ruiz.
- D) Ruiz is best known for works about migration.

Answer: A

Question 25

Skill: Expression of Ideas: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- The Sola Canal opened in 1895.
- The canal shortened travel between two inland cities.
- Before the canal opened, goods were transported by road over the eastern hills.
- Transporting goods by canal reduced travel time by two days.

The student wants to explain the canal's practical importance. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The Sola Canal shortened travel between two inland cities and reduced transport time by two days.
- B) The Sola Canal opened in 1895.
- C) Before the Sola Canal opened, goods were transported by road.
- D) The eastern hills were used as a trade route before 1895.

Answer: A

Question 26

Skill: Expression of Ideas: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- Botanist Keiko Arai studied alpine mosses.
- Arai's 2011 paper focused on mosses that survive in freezing temperatures.
- Arai found that some alpine mosses produce protective sugars during cold months.
- These sugars help prevent cell damage.

The student wants to present Arai's main finding. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Arai found that some alpine mosses produce protective sugars that help prevent cell damage during cold months.
- B) Keiko Arai studied alpine mosses.
- C) Arai published a paper in 2011.
- D) Some alpine mosses survive in freezing temperatures.

Answer: A

Question 27

Skill: Expression of Ideas: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- Composer Idris Hale wrote music for silent films.
- Hale often used short, repeated melodies called motifs.
- In the 1924 film *The Lighthouse Road*, a three-note motif plays whenever the sea appears.
- Critics say this motif creates unity across the film.

The student wants to give a specific example of Hale's use of motifs. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) In *The Lighthouse Road*, a three-note motif plays whenever the sea appears.
- B) Composer Idris Hale wrote music for silent films.
- C) Motifs are short, repeated melodies.
- D) Critics say one motif creates unity across *The Lighthouse Road*.

Answer: A

SECTION 2, MODULE 1: Math

Question 1

Skill: Linear Equations in One Variable

If $5x - 7 = 28$, what is the value of x ?

- A) 5
- B) 6
- C) 7
- D) 8

Answer: C

Solution: Add 7 to both sides: $5x = 35$. Divide by 5 to get $x = 7$.

Question 2

Skill: Ratios and Proportions

A recipe uses 3 cups of oats for every 2 cups of dried fruit. If 12 cups of oats are used, how many cups of dried fruit are needed?

- A) 6
- B) 8
- C) 10
- D) 18

Answer: B

Solution: The oat amount is multiplied by 4, so the dried fruit amount is also multiplied by 4: $2 \times 4 = 8$.

Question 3

Skill: Percentages

A jacket originally costs \$80. During a sale, its price is reduced by 25%. What is the sale price of the jacket?

- A) \$20
- B) \$55
- C) \$60
- D) \$75

Answer: C

Solution: 25% of 80 is 20, so the sale price is $80 - 20 = 60$ dollars.

Question 4

Skill: Systems of Linear Equations

A system of equations is shown:

$$2x + y = 13$$

$$x - y = 2$$

What is the value of x ?

- A) 3
- B) 4
- C) 5
- D) 6

Answer: C

Solution: Add the equations: $3x = 15$, so $x = 5$.

Question 5

Skill: Linear Functions

The function f is defined by $f(x) = 4x + 9$. What is the value of $f(6)$?

- A) 24
- B) 30
- C) 33
- D) 39

Answer: C

Solution: Substitute 6 for x : $f(6) = 4(6) + 9 = 24 + 9 = 33$.

Question 6

Skill: Geometry: Area

A rectangle has length 14 centimeters and width 9 centimeters. What is its area, in square centimeters?

Answer: 126

Solution: Area of a rectangle is length times width: $14 \times 9 = 126$.

Question 7

Skill: Quadratic Equations

What are the solutions to $x^2 - 9x + 20 = 0$?

- A) 2 and 10
- B) 4 and 5
- C) -4 and -5
- D) -2 and -10

Answer: B

Solution: Factor the quadratic: $x^2 - 9x + 20 = (x - 4)(x - 5)$. Thus $x = 4$ or $x = 5$.

Question 8

Skill: Exponents

If $2^a = 32$, what is the value of a ?

- A) 4
- B) 5
- C) 6
- D) 16

Answer: B

Solution: $32 = 2^5$, so $a = 5$.

Question 9

Skill: Data Analysis

The numbers of books read by five students are 3, 7, 8, 10, and 12. What is the mean number of books read?

Answer: 8

Solution: The sum is $3 + 7 + 8 + 10 + 12 = 40$. Divide by 5 to get 8.

Question 10

Skill: Linear Equations in Two Variables

The line $y = mx + b$ passes through the points (0, 5) and (4, 17). What is the slope m of the line?

- A) 2
- B) 3
- C) 4
- D) 5

Answer: B

Solution: Slope is $(17 - 5)/(4 - 0) = 12/4 = 3$.

Question 11

Skill: Geometry: Angles

Two angles are supplementary. One angle measures 112 degrees. What is the measure, in degrees, of the other angle?

Answer: 68

Solution: Supplementary angles sum to 180 degrees. $180 - 112 = 68$.

Question 12

Skill: Function Notation

If $g(x) = x^2 - 3x$, what is $g(-2)$?

- A) -10
- B) -2
- C) 2
- D) 10

Answer: D

Solution: $g(-2) = (-2)^2 - 3(-2) = 4 + 6 = 10$.

Question 13

Skill: Linear Inequalities

Which of the following values of x satisfies $3x + 4 > 19$?

- A) 3
- B) 4
- C) 5
- D) 6

Answer: D

Solution: $3x + 4 > 19$ gives $3x > 15$, so $x > 5$. Only 6 is greater than 5.

Question 14

Skill: Equivalent Expressions

Which expression is equivalent to $6(2x - 5) + 3x$?

- A) $9x - 30$
- B) $12x - 2$
- C) $15x - 30$
- D) $15x - 5$

Answer: C

Solution: Distribute: $6(2x - 5) + 3x = 12x - 30 + 3x = 15x - 30$.

Question 15

Skill: Probability

A bag contains 5 red marbles and 7 blue marbles. If one marble is selected at random, what is the probability that it is blue?

- A) $5/12$
- B) $7/12$
- C) $5/7$
- D) $7/5$

Answer: B

Solution: There are 12 marbles total and 7 are blue, so the probability is $7/12$.

Question 16

Skill: Geometry: Volume

A rectangular prism has length 5 inches, width 4 inches, and height 9 inches. What is its volume, in cubic inches?

Answer: 180

Solution: Volume is length times width times height: $5 \times 4 \times 9 = 180$.

Question 17

Skill: Rates

A printer prints 36 pages in 4 minutes. At this rate, how many pages does it print in 9 minutes?

- A) 72
- B) 81
- C) 90
- D) 108

Answer: B

Solution: The printer prints $36/4 = 9$ pages per minute. In 9 minutes, it prints $9 \times 9 = 81$ pages.

Question 18

Skill: Coordinate Geometry

What is the distance between the points (2, 3) and (2, 11) in the xy-plane?

- A) 6
- B) 8
- C) 10
- D) 14

Answer: B

Solution: The x-coordinates are the same, so the vertical distance is $|11 - 3| = 8$.

Question 19

Skill: Algebra: Solving for a Variable

If $4p + 3q = 25$ and $p = 4$, what is the value of q ?

Answer: 3

Solution: Substitute $p = 4$: $4(4) + 3q = 25$, so $16 + 3q = 25$. Then $3q = 9$ and $q = 3$.

Question 20

Skill: Quadratic Functions

The function h is defined by $h(x) = (x - 3)^2 + 2$. What is the minimum value of $h(x)$?

- A) -3
- B) 2
- C) 3
- D) 5

Answer: B

Solution: The expression $(x - 3)^2$ is always at least 0, so the minimum value is $0 + 2 = 2$.

Question 21

Skill: Statistics

The data set 4, 6, 8, 10, 12 has a median of 8. If 20 is added to the data set, what is the median of the new data set?

- A) 8
- B) 9
- C) 10
- D) 11

Answer: B

Solution: The ordered set is 4, 6, 8, 10, 12, 20. The median is the average of the 3rd and 4th values: $(8 + 10)/2 = 9$.

Question 22

Skill: Nonlinear Equations

If $x^2 = 49$ and x is positive, what is the value of x ?

Answer: 7

Solution: The positive number whose square is 49 is 7.

SECTION 2, MODULE 2: Math

Question 1

Skill: Linear Equations in One Variable

If $7(2x - 3) = 63$, what is the value of x ?

- A) 4
- B) 5
- C) 6
- D) 7

Answer: C

Solution: Divide by 7: $2x - 3 = 9$. Add 3: $2x = 12$. Divide by 2 to get $x = 6$.

Question 2

Skill: Percentages

The population of a town increased from 12,000 to 15,000. By what percent did the population increase?

- A) 20%
- B) 25%
- C) 30%
- D) 35%

Answer: B

Solution: The increase is 3,000. Since $3,000/12,000 = 0.25$, the percent increase is 25%.

Question 3

Skill: Systems of Linear Equations

A ticket booth sold adult tickets for \$12 each and student tickets for \$8 each. On one day, 40 tickets were sold for a total of \$400. How many adult tickets were sold?

Answer: 20

Solution: Let a be adult tickets and s be student tickets. $a + s = 40$ and $12a + 8s = 400$. Substitute $s = 40 - a$: $12a + 8(40 - a) = 400$, so $4a + 320 = 400$ and $a = 20$.

Question 4

Skill: Quadratic Equations

Which of the following is a solution to $x^2 + 6x = 16$?

- A) -8
- B) -4
- C) 2
- D) 8

Answer: C

Solution: Rewrite as $x^2 + 6x - 16 = 0$. Factor: $(x + 8)(x - 2) = 0$, so $x = -8$ or $x = 2$. From the options, 2 is a solution.

Question 5

Skill: Exponential Functions

A bacteria culture starts with 500 bacteria and doubles every hour. Which expression gives the number of bacteria after t hours?

- A) $500(2)^t$
- B) $2(500)^t$
- C) $500 + 2t$
- D) $500t^2$

Answer: A

Solution: Doubling every hour means multiply by 2 once for each hour, so the model is $500(2)^t$.

Question 6

Skill: Geometry: Circles

A circle has radius 6. What is its area in terms of π ?

- A) 6π
- B) 12π
- C) 36π
- D) 72π

Answer: C

Solution: Area of a circle is πr^2 . With $r = 6$, the area is 36π .

Question 7

Skill: Linear Functions

A line has slope -2 and passes through the point (3, 10). What is the y-intercept of the line?

Answer: 16

Solution: Use $y = mx + b$. Substitute $m = -2$ and $(x, y) = (3, 10)$: $10 = -2(3) + b$, so $10 = -6 + b$ and $b = 16$.

Question 8

Skill: Equivalent Expressions

For x not equal to 0, which expression is equivalent to $(12x^3 - 8x^2)/(4x^2)$?

- A) $3x - 2$
- B) $3x^2 - 2$
- C) $8x$
- D) $12x - 8$

Answer: A

Solution: Divide each term by $4x^2$: $12x^3/(4x^2) = 3x$ and $-8x^2/(4x^2) = -2$. The result is $3x - 2$.

Question 9

Skill: Geometry: Similar Triangles

Two similar triangles have corresponding side lengths 5 and 15. The area of the smaller triangle is 12 square units. What is the area of the larger triangle?

- A) 36
- B) 72
- C) 108
- D) 180

Answer: C

Solution: The scale factor for length is $15/5 = 3$. Areas scale by the square of the length scale factor: $3^2 = 9$. The larger area is $12 \times 9 = 108$.

Question 10

Skill: Statistics

A data set has mean 18. If each value in the data set is increased by 5, what is the mean of the new data set?

Answer: 23

Solution: Adding 5 to every value increases the mean by 5. The new mean is $18 + 5 = 23$.

Question 11

Skill: Linear Inequalities

A shipping company charges a flat fee of \$15 plus \$4 per package. What is the greatest number of packages that can be shipped for at most \$63?

- A) 10
- B) 11
- C) 12
- D) 13

Answer: C

Solution: Let p be the number of packages. $15 + 4p \leq 63$, so $4p \leq 48$ and $p \leq 12$. The greatest integer is 12.

Question 12

Skill: Function Transformations

The function f is defined by $f(x) = x^2$. The function g is defined by $g(x) = f(x - 4) + 1$. What is $g(6)$?

- A) 3
- B) 5
- C) 9
- D) 37

Answer: B

Solution: $g(6) = f(6 - 4) + 1 = f(2) + 1 = 2^2 + 1 = 5$.

Question 13

Skill: Ratios

In a mixture, the ratio of vinegar to oil is 2 to 5. If the mixture contains 35 milliliters of oil, how many milliliters of vinegar does it contain?

Answer: 14

Solution: Oil corresponds to 5 parts, so each part is $35/5 = 7$ milliliters. Vinegar is 2 parts, so $2 \times 7 = 14$ milliliters.

Question 14

Skill: Advanced Algebra: Rational Equations

For x not equal to -3 , if $(x^2 + 5x + 6)/(x + 3) = 6$, what is x ?

- A) 3
- B) 4
- C) 6
- D) 9

Answer: B

Solution: Factor the numerator: $x^2 + 5x + 6 = (x + 2)(x + 3)$. Since x is not -3 , the expression simplifies to $x + 2$. Then $x + 2 = 6$, so $x = 4$.

Question 15

Skill: Geometry: Right Triangles

A right triangle has legs of lengths 9 and 12. What is the length of the hypotenuse?

- A) 15
- B) 18
- C) 21
- D) 25

Answer: A

Solution: By the Pythagorean theorem, $c^2 = 9^2 + 12^2 = 81 + 144 = 225$, so $c = 15$.

Question 16

Skill: Quadratic Functions

The graph of $y = x^2 - 10x + 27$ has a minimum point. What is the y -coordinate of that point?

Answer: 2

Solution: The x -coordinate of the vertex is $-b/(2a) = 10/2 = 5$. Substitute: $5^2 - 10(5) + 27 = 25 - 50 + 27 = 2$.

Question 17

Skill: Data Analysis

A scatterplot shows a positive linear association between the number of hours a student studies and the student's test score. Which statement best describes this association?

- A) Students who study more hours tend to have higher test scores.
- B) Students who study more hours tend to have lower test scores.
- C) Studying time and test score have no relationship.
- D) Every student who studies the same number of hours has exactly the same score.

Answer: A

Solution: A positive association means that as one variable increases, the other tends to increase.

Question 18

Skill: Trigonometry

In a right triangle, an acute angle has opposite side length 7 and hypotenuse length 25. What is the sine of the angle?

- A) $7/25$
- B) $24/25$
- C) $7/24$
- D) $24/7$

Answer: A

Solution: Sine equals opposite divided by hypotenuse, so $\sin(\theta) = 7/25$.

Question 19

Skill: Linear Models

A gym charges a one-time registration fee plus a monthly fee. The total cost for 3 months is \$145, and the total cost for 7 months is \$285. What is the monthly fee?

Answer: 35

Solution: The cost increases by $285 - 145 = 140$ dollars over $7 - 3 = 4$ months. The monthly fee is $140/4 = 35$ dollars.

Question 20

Skill: Polynomial Expressions

If $(x + 2)(x + k) = x^2 + 11x + 18$, what is the value of k ?

- A) 6
- B) 7
- C) 8
- D) 9

Answer: D

Solution: Expanding gives $x^2 + (k + 2)x + 2k$. Since the constant term is 18, $2k = 18$, so $k = 9$. Then $k + 2 = 11$, which matches.

Question 21

Skill: Probability

A number is selected at random from the integers 1 through 30, inclusive. What is the probability that the number is a multiple of 4?

- A) $1/5$
- B) $7/30$
- C) $1/4$
- D) $2/5$

Answer: B

Solution: The multiples of 4 from 1 to 30 are 4, 8, 12, 16, 20, 24, and 28, for 7 numbers. The probability is $7/30$.

Question 22

Skill: Exponents and Radicals

If $\sqrt{3x + 1} = 5$, what is the value of x ?

Answer: 8

Solution: Square both sides to get $3x + 1 = 25$. Then $3x = 24$, so $x = 8$.

Answer Key

SECTION 1, MODULE 1: Reading and Writing

1. A 2. C 3. A 4. A 5. A 6. A 7. A 8. A 9. A
10. A 11. A 12. A 13. A 14. A 15. A 16. A 17. A 18. B
19. A 20. A 21. A 22. A 23. A 24. A 25. A 26. A 27. A

SECTION 1, MODULE 2: Reading and Writing

1. A 2. A 3. A 4. A 5. A 6. A 7. A 8. A 9. A
10. A 11. A 12. A 13. A 14. A 15. A 16. A 17. A 18. A
19. A 20. A 21. A 22. A 23. A 24. A 25. A 26. A 27. A

SECTION 2, MODULE 1: Math

1. C 2. B 3. C 4. C 5. C 6. 126 7. B 8. B
9. 8 10. B 11. 68 12. D 13. D 14. C 15. B 16. 180
17. B 18. B 19. 3 20. B 21. B 22. 7

SECTION 2, MODULE 2: Math

1. C 2. B 3. 20 4. C 5. A 6. C 7. 16 8. A
9. C 10. 23 11. C 12. B 13. 14 14. B 15. A 16. 2
17. A 18. A 19. 35 20. D 21. B 22. 8